Transfer learning

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

If two different records become the same feature vector, can a deterministic output head receiving only that vector distinguish them?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Trace a reused feature extractor and a newly fitted output head. Try the recall before opening notes.

Your learning target

I can trace a reused **feature extractor** and a newly fitted **output head**.

I can fit a small target-task head using **training** data and compare a baseline.

I can identify when a frozen representation cannot support a new task.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Trace a reused **feature extractor** and a newly fitted **output head**

Fit a small target-task head using **training** data and compare a baseline

Identify when a frozen representation cannot support a new task

Retrieve what you already know

> If two different records become the same feature vector, can a deterministic **output head** receiving only that vector distinguish them?

My starting explanation

This week's end goal

By the end, I can explain **transfer learning** using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Transfer learning case. Separate observations, given facts and assumptions before choosing an explanation.

A source model may learn useful features before we have many labels for a new target task. **Transfer learning** reuses those learned features or parameters. One approach freezes the **feature extractor** and fits a new **output head**.

Fine tuning instead updates some or all of the reused parameters, usually together with the head.

Use a transparent stand-in for a source-trained extractor: f(x) = ReLU(x1 + x2), with supplied weights [1, 1] and bias 0. ReLU keeps a positive value and replaces a negative value with zero.

Here x1 and x2 are nonnegative fictional event counts, so f(x) is their sum. We did not train this extractor on a source dataset; this exercise demonstrates the mechanics of reuse, not measured transfer benefits.

Part	Operation	During this exercise
Input	[count of type 1, count of type 2]	Keep order and units
Extractor	h = max(0, x1 + x2)	Freeze weights and bias
New head	Predict 1 if h >= threshold	Fit the threshold on target training data

The target is a high-activity label. The extractor returns a single number h, not a class label or probability. A new head converts h into the target **prediction**.

List the given facts and one fact that is not supplied.

See how it works

> Follow the mechanism for Transfer learning. Connect each labelled input, deciding step and result.

Use target **training** records T1 [1, 1] with label 0, T2 [1, 3] with label 0, T3 [4, 2] with label 1 and T4 [5, 3] with label 1. Their extracted values are 2, 4, 6 and 8.

Fit the head by trying thresholds 3, 5 and 7 and counting **training** mistakes. **Fitting** can use a small search; it need not use gradients.

```
train = [([1, 1], 0), ([1, 3], 0),
         ([4, 2], 1), ([5, 3], 1)]

def feature(x):
    if len(x) != 2 or any(v < 0 for v in x):
        raise ValueError("Need two nonnegative counts")
    return max(0, x[0] + x[1])

def predict(x, threshold):
    return int(feature(x) >= threshold)

def mistakes(threshold):
    return sum(predict(x, threshold) != y
               for x, y in train)

candidates = [3, 5, 7]
best = min(candidates, key=lambda t: (mistakes(t), t))
assert best == 5
print([mistakes(t) for t in candidates])  # [1, 0, 1]
```

The tie rule prefers the smaller threshold if **training** mistake counts match; there is no tie here. The extractor remains unchanged. The code assumes finite numeric counts and valid binary labels.

A constant baseline predicts the most frequent **training** label; with two of each, our declared tie rule chooses 0.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Transfer learning. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

For the four target **training** records, give the number of mistakes at thresholds 3, 5 and 7 and choose the fitted threshold. Then score that head and the declared constant baseline on V1 and V2.

Explain which data fitted the head.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> **Change one thing**
>
> Change: Query changes to [2, 1].

Predicted result

Prediction: Its h becomes 4 and the fitted head predicts class 0.

Why the result follows

Reason: Extractor stays fixed; the changed **input** crosses the chosen head's boundary.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Transfer learning to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Transfer learning	Reusing knowledge learned on a source task or dataset to help a target task
Feature extractor	The part of a model that maps an input to a representation
Output head	The part that maps a representation to the current task's output
Fine tuning	Training some or all previously learned parameters on the target task

Use two terms to explain a step in this week's example.

Follow a worked example

Follow the worked trace for Transfer learning, predict a step, then compare with its explanation.

Freeze the fitted head before validation

VALIDATION INPUT	FROZEN FEATURE	HEAD AT THRESHOLD 5
V1 [2, 1], label 0	h = 3	Predict 0, correct
V2 [3, 4], label 1	h = 7	Predict 1, correct

The fitted head is correct on 2 of 2 validation records; the constant -0 baseline is correct on 1 of 2. This tiny constructed comparison is not evidence of general transfer performance.

Worked explanation · Transfer learning

Use the same validation records for both methods. These records were not used to fit the threshold. If validation guides later choices, reserve separate untouched final evaluation data.

We have not compared with a full model trained from scratch or measured **training** time, so this result cannot establish either advantage.

A new input [2, 3] uses the fitted threshold 5. Give h and the predicted class, including the equality rule. Which parameters changed during fitting?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Transfer learning

Transfer a fixed **feature extractor** while identifying what is actually refitted.

The facts we will use

Supplied facts

Frozen extractor h =left+2right. New-task **training input**s [1, 0]label 0,[2, 1]label 0,[1, 3]label 1. Class 1 when h >=threshold; candidates 3 or5.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/28

Transfer learning

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Extract features

Training h values are 1, 4, 7.



2. Fit new head

Threshold 3 makes 1 mistake; threshold 5 makes 0, so choose 5.



3. Check held-out input

Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

CHANGE ONE THING

Query changes to [2, 1].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Extract features	Training h values are 1, 4, 7.
2. Fit new head	Threshold 3 makes 1 mistake; threshold 5 makes 0, so choose 5.
3. Check held-out input	Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Transfer learning can keep a representation fixed while **fitting** a smaller **output head** for a new task. Here the extractor always applies the same weights, while new labels choose the head's threshold.

Applying the chosen head to a query is inference. State which part was frozen and which was fitted; this arithmetic does not establish the extractor is useful for every task.

What this example does not establish

Keep the stated limit

A frozen extractor can miss task-specific distinctions; transfer is a strategy requiring

validation, not a guarantee of useful features.

Keep the stated limit

Read the labelled picture: Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Query changes to [2, 1]. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Transfer learning claim and explain a repair using the supplied evidence.

Check the claim against the facts

Now change the target: predict 1 when the first count is greater than the second, otherwise 0. U [1, 3] has label 0 and V [3, 1] has label 1, but both become h = 4.

Every fixed deterministic head receiving only this h must give both the same **prediction**. At least one of the two will be wrong, regardless of how many extra head parameters we add.

Check the claim against the facts

Fine tuning could change the extractor to retain relevant distinctions; it does not guarantee success. To see a possible change, hand-set the weights to [1, -1], keep bias 0 and use threshold 1.

U then gives ReLU(-2) = 0 and predicts 0; V gives ReLU(2) = 2 and predicts 1. This hand-chosen repair is not a trained result or an independent evaluation.

Keeping the original **input** alongside h is another possible representation change. Any such change needs a new **training** and evaluation plan.

Normalising the **input**s also changes the pipeline; last week showed why that can remove useful magnitude information.

Locate the earliest unsupported step

For the first-count-greater target, explain why the original extractor cannot distinguish U [1, 3] from V [3, 1]. Trace the hand-chosen [1, -1] extractor and threshold 1.

Does this show that **fine tuning** has been trained and independently validated?

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Transfer learning with support, then explain the deciding step in your own words.

For **input** [4, 2], compute h using the original extractor and the class at threshold 5. Identify the frozen weights and bias, the fitted quantity, and why h is not a probability.

> **A new input [2, 3] uses the fitted threshold 5. Give h and the predicted class, including the equality rule. Which parameters changed during fitting?**

> **Explain your prediction to a partner before comparing results.**

Practise with less help: Fit and compare

Use workbook W2 for the connected example "**Transfer learning**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Transfer learning**. Proposed explanation: "Frozen means the whole model cannot learn". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Transfer learning. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

For the four target **training** records, give the number of mistakes at thresholds 3, 5 and 7 and choose the fitted threshold. Then score that head and the declared constant baseline on V1 and V2.

Explain which data fitted the head.

```
train = [([1, 1], 0), ([1, 3], 0),
         ([4, 2], 1), ([5, 3], 1)]

def feature(x):
    if len(x) != 2 or any(v < 0 for v in x):
        raise ValueError("Need two nonnegative counts")
    return max(0, x[0] + x[1])
```

```
def predict(x, threshold):
    return int(feature(x) >= threshold)

def mistakes(threshold):
    return sum(predict(x, threshold) != y
               for x, y in train)

candidates = [3, 5, 7]
best = min(candidates, key=lambda t: (mistakes(t), t))
assert best == 5
print([mistakes(t) for t in candidates])  # [1, 0, 1]
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Transfer learning. Explain why each step

> is justified.

Explain → apply → challenge

For **input** [4, 2], compute h using the original extractor and the class at threshold 5. Identify the frozen weights and bias, the fitted quantity, and why h is not a probability.

> For the first-count-greater target, explain why the original extractor cannot distinguish U [1, 3] from V [3, 1]. Trace the hand-chosen [1, -1] extractor and threshold 1.
>
> Does this show that **fine tuning** has been trained and independently validated?

A report omits the source checkpoint and pre**process**ing, claims faster learning without timing, and repeatedly chooses settings using the final evaluation set.

Name the missing records and propose a fair comparison with a model trained from scratch.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **transfer learning evidence** sheet to Image-pattern Network Investigation. Show how you can trace a reused **feature extractor** and a newly fitted **output head**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

For a real source-trained model, record the checkpoint version, source task and data provenance, permitted use, **input** order and pre**process**ing, frozen and trainable parts, target splits, baseline and resource budget.

Check for source-data overlap with target evaluation records; a different filename is not proof of independence. If source history is unknown, report that limitation.

A team says its reused extractor must outperform training from scratch because it scored 2 of 2 here. Give two reasons this conclusion is unsupported and propose a fair next comparison.

Evaluate methods on the same target examples under a declared **training** budget, with choices made before final evaluation. Record errors as well as totals.

Next week examines uncertainty and failure analysis, including confident **prediction**s that can still be wrong.

A report omits the source checkpoint and pre**process**ing, claims faster learning without timing, and repeatedly chooses settings using the final evaluation set.

Name the missing records and propose a fair comparison with a model trained from scratch.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Transfer learning case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Freeze the fitted head before validation

VALIDATION INPUT	FROZEN FEATURE	HEAD AT THRESHOLD 5
V1 [2, 1], label 0	h = 3	Predict 0, correct
V2 [3, 4], label 1	h = 7	Predict 1, correct

The fitted head is correct on 2 of 2 validation records; the constant -0 baseline is correct on 1 of 2. This tiny constructed comparison is not evidence of general transfer performance.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Fit a new head independently

A separate frozen extractor is h = 2*x1 + x2. **Training input**s [1, 0], [1, 2], [2, 1], [3, 1] have labels 0, 0, 1, 1.

Compare thresholds 2, 4 and 6 using **prediction** 1 when h >= threshold; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation [2, 0] with label 1 and [0, 2] with label 0.

Compare with the **training**-majority baseline, whose tie rule chooses 0.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A separate frozen extractor is h = 2*x1 + x2. **Training input**s [1, 0], [1, 2], [2, 1], [3, 1] have labels 0, 0, 1, 1.

Compare thresholds 2, 4 and 6 using **prediction** 1 when h >= threshold; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation [2, 0] with label 1 and [0, 2] with label 0.

Compare with the **training**-majority baseline, whose tie rule chooses 0.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Transfer learning, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Transfer a fixed **feature extractor** while identifying what is actually refitted.

By the end, I can	How I will show it
Trace a reused feature extractor and a newly fitted output head	Explain the deciding step, show a trace or test, and state one limit.
Fit a small target-task head using training data and compare a baseline	Explain the deciding step, show a trace or test, and state one limit.
Identify when a frozen representation cannot support a new task	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Transfer learning without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Trace a reused **feature extractor** and a newly fitted **output head**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Transfer a fixed **feature extractor** while identifying what is actually refitted.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

1. Check held-out **input**

2. Extract features

3. Fit new head

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check held-out **input** Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

Transfer learning can keep a representation fixed while **fitting** a smaller **output head** for a new task. Here the extractor always applies the same weights, while new labels choose the head's threshold.

Applying the chosen head to a query is inference. State which part was frozen and which was fitted; this arithmetic does not establish the extractor is useful for every task.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Query changes to [2, 1]. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Its h becomes 4 and the fitted head predicts class 0.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Its h becomes 4 and the fitted head predicts class 0. Extractor stays fixed; the changed **input** crosses the chosen head's boundary.

Repair a missing building block

Review Similarity and normalisation (ai-5/27). Goal: Calculate vector length and normalise a nonzero vector. Short practice: Why do the original two metrics select different winners?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sequence representations (ai-5/25). Goal: Encode an ordered sequence with an explicit vocabulary. Short practice: Which record lets the program distinguish the first and last zeros?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

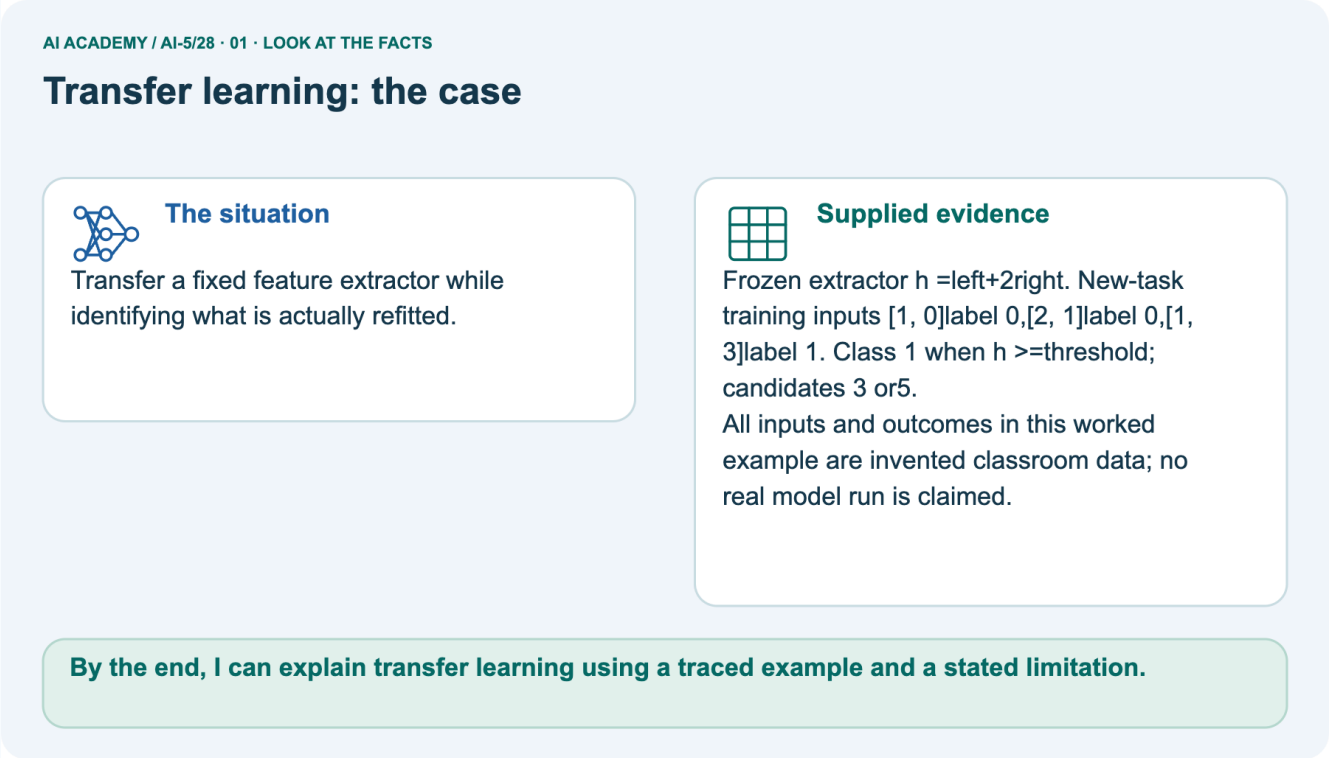
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Transfer learning graphic using its labels, deciding evidence and relevant limitation.

01. Transfer learning: the case



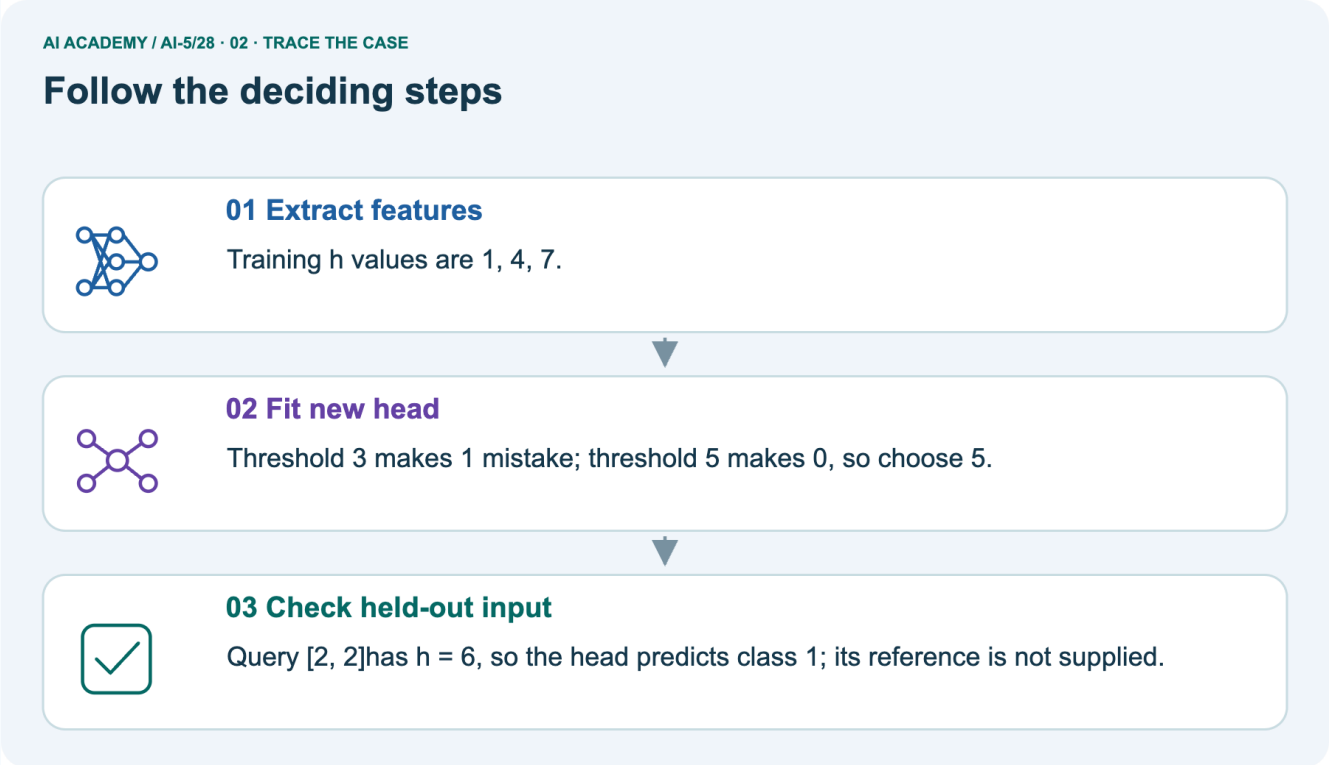
Follow the labelled picture with the matching explanation. Open the vector diagram

Transfer a fixed **feature extractor** while identifying what is actually refitted. Supplied facts: Frozen extractor h =left+2right. New-task **training input**s [1, 0]label 0,[2, 1]label 0,[1, 3]label 1.

Class 1 when h >=threshold; candidates 3 or5. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

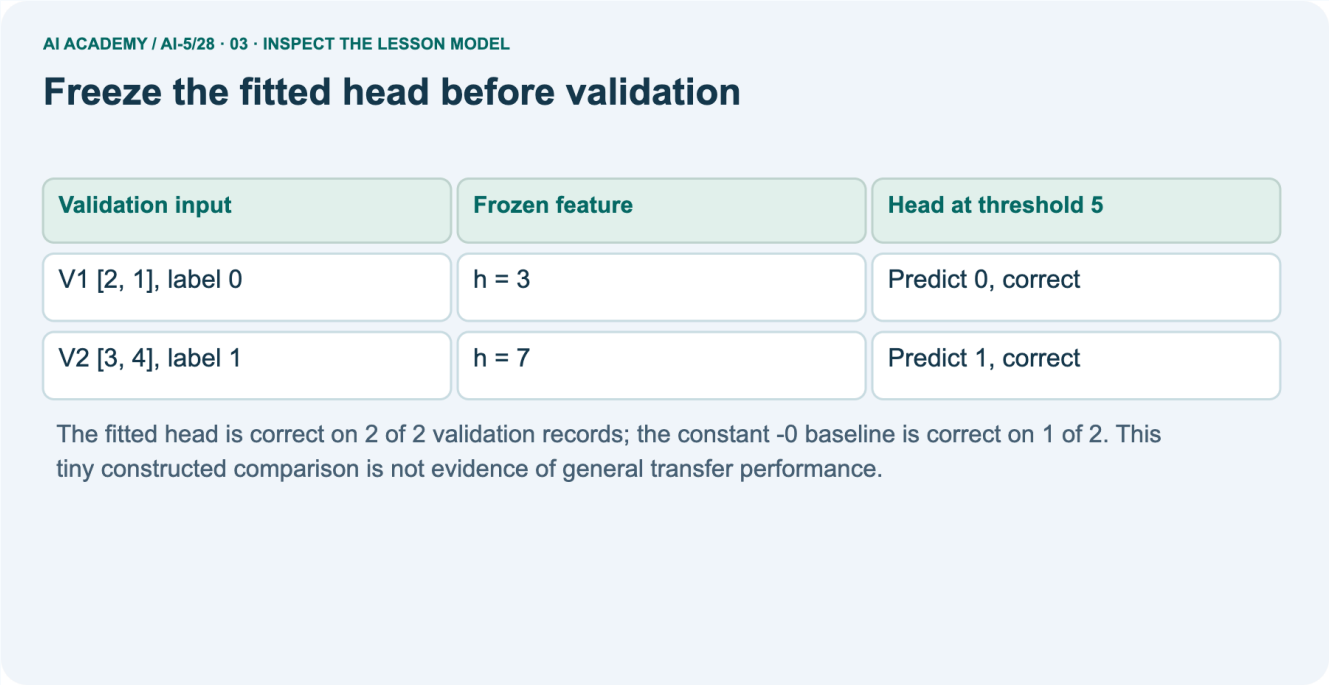


Follow the labelled picture with the matching explanation. Open the vector diagram

Extract features: **Training** h values are 1, 4, 7. → Fit new head: Threshold 3 makes 1 mistake; threshold 5 makes 0, so choose 5. → Check held-out **input**: Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

Point to each numbered stage and explain which recorded **input** it uses.

03. Freeze the fitted head before validation

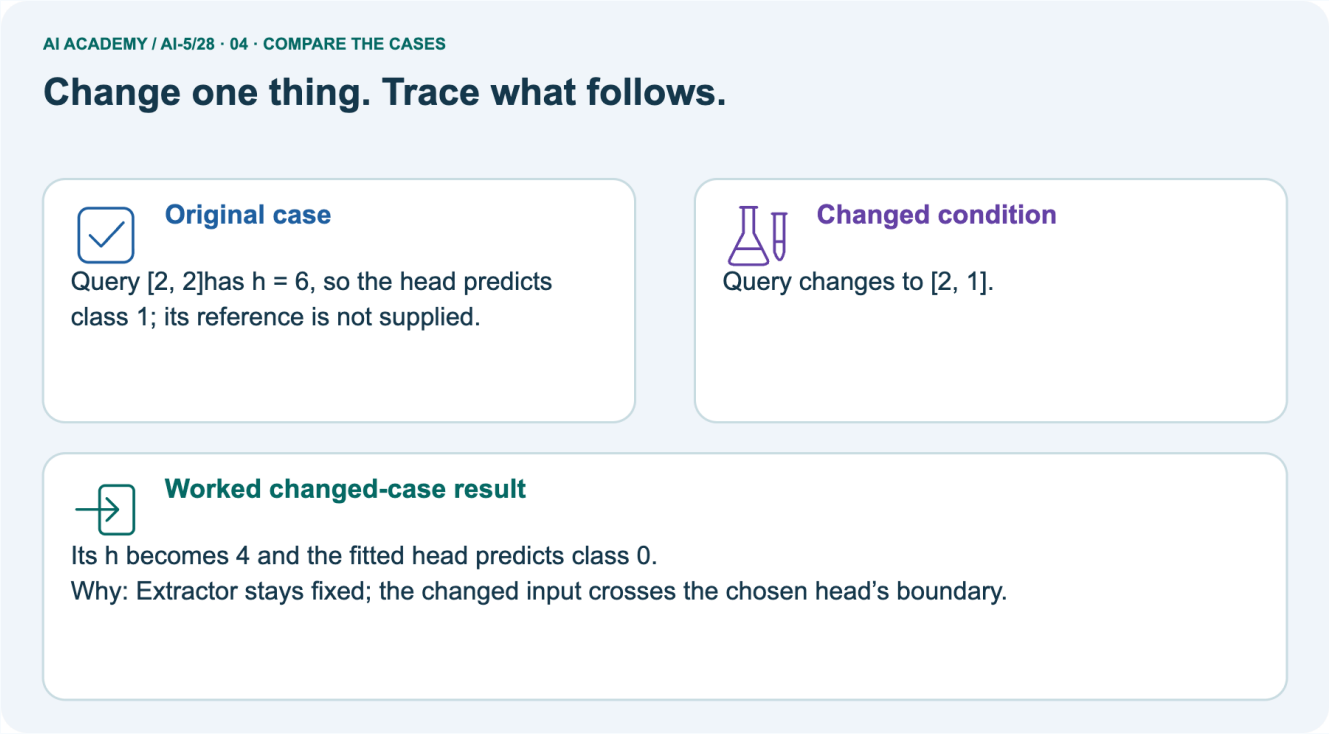


Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze the fitted head before validation. V1 [2, 1], label 0 | h = 3 | Predict 0, correct. V2 [3, 4], label 1 | h = 7 | Predict 1, correct

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



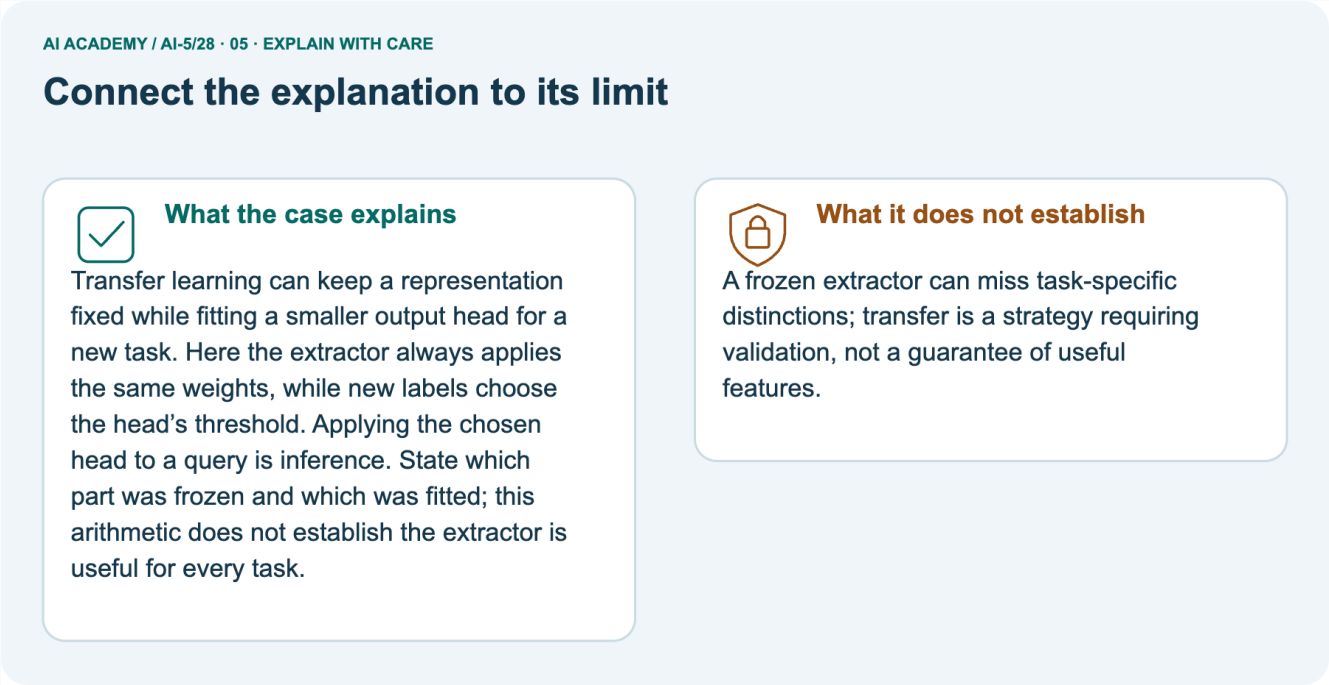
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied. Changed condition: Query changes to [2, 1]. Expected result: Its h becomes 4 and the fitted head predicts class 0.

Reason: Extractor stays fixed; the changed **input** crosses the chosen head’s boundary.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

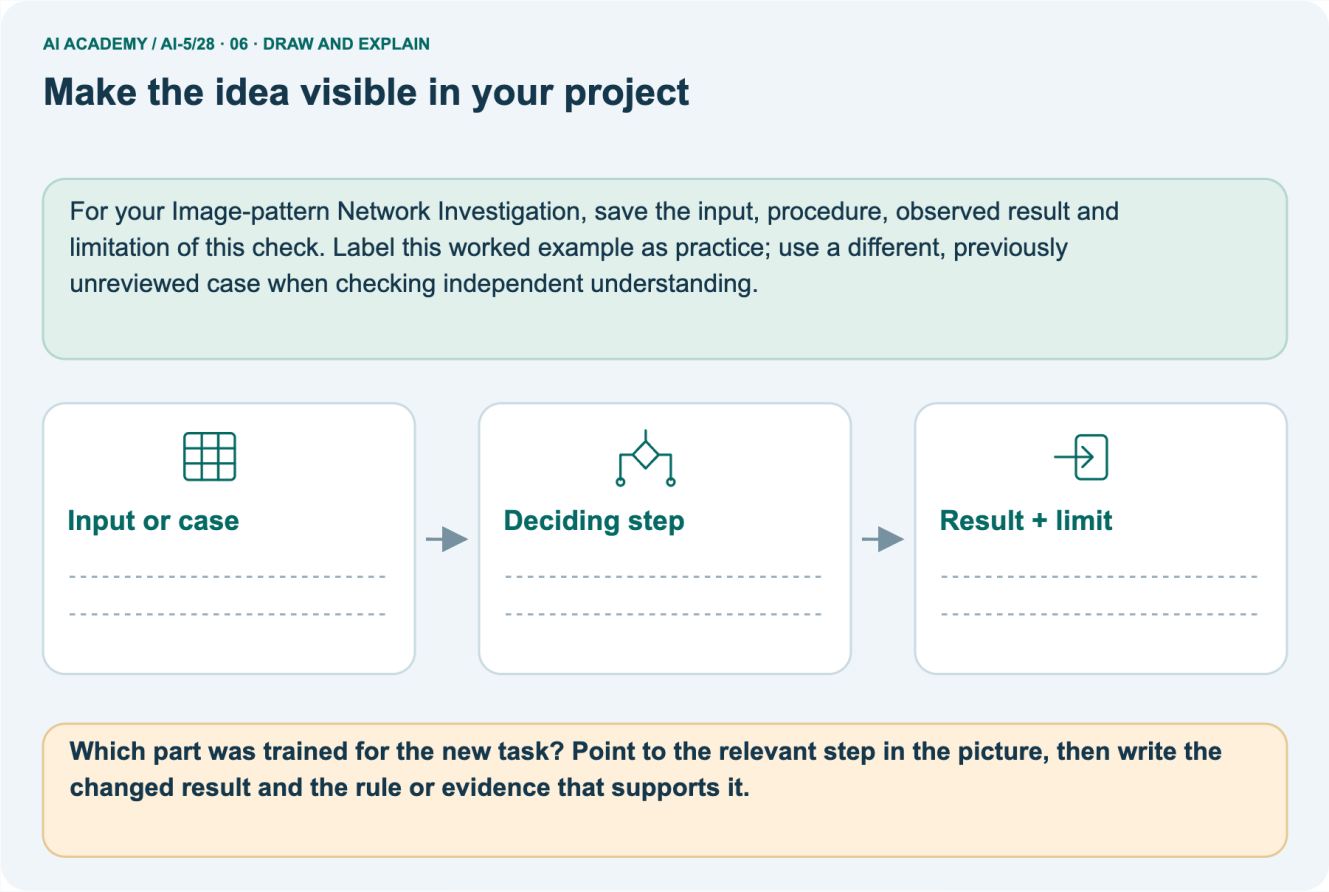
Transfer learning can keep a representation fixed while **fitting** a smaller **output head** for a new task. Here the extractor always applies the same weights, while new labels choose the head's threshold.

Applying the chosen head to a query is inference. State which part was frozen and which was fitted; this arithmetic does not establish the extractor is useful for every task.

Limit: A frozen extractor can miss task-specific distinctions; transfer is a strategy requiring validation, not a guarantee of useful features.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.